

Table 1. Fine-tuning Phi on synthetic examples generated by GRADMM, vs LLM-generated zero-shot and few-shot synthetic data, vs real examples selected with herding, K-center, and Random baselines. Synthetic data generated by GRADMM outperforms the baselines by up to 10.4% and is the only method that can preserve the privacy of the training data. GRADMM’s synthetic data has similar log-perplexity (ppl) to that of real data, and higher ppl than LLM-generated synthetic data, confirming its more diverse nature.

		Privacy $\checkmark$			LLM generated, no privacy X				Real data, no privacy X						
Dataset	# data	GRADMM			Zero-shot		Few-shot		Herding		K-center		Random		Rand 1K
		$\epsilon = \infty$	$\epsilon = 0.05$	ppl	acc	ppl	acc	ppl	acc	ppl	acc	ppl	acc	ppl	
SST-2	5	86.5 ± 0.5	84.2 ± 0.1	5.8	71.6 ± 4.4	2.5	71.8 ± 0.7	3.0	61.2 ± 5.7	6.6	75.3 ± 0.9	5.5	52.9 ± 2.4	7.7	91.3 ± 0.4
	10	87.4 ± 0.9	86.2 ± 1.8	5.1	79.1 ± 6.5	2.3	72.0 ± 0.3	2.6	78.8 ± 0.9	6.7	82.0 ± 1.0	5.5	66.1 ± 11.5	7.3	
	20	87.9 ± 0.3	87.2 ± 0.8	5.3	82.2 ± 2.2	2.2	77.5 ± 0.5	2.7	83.3 ± 2.8	6.6	86.2 ± 0.9	5.7	86.2 ± 1.1	7.0	
	Base acc:	89.7 ± 0.2	88.0 ± 0.1	5.7	83.0 ± 1.1	2.3	80.6 ± 3.9	2.6	86.0 ± 0.4	6.6	88.1 ± 1.0	5.5	87.8 ± 0.9	6.8	
	69.6%	89.7 ± 0.1	88.6 ± 0.5	5.2	87.5 ± 0.6	2.3	77.4 ± 1.7	2.5	88.9 ± 0.1	6.7	89.3 ± 0.5	5.8	88.8 ± 0.8	6.7	
Tweet emotions	5	83.8 ± 0.3	80.4 ± 1.7	3.5	70.7 ± 10.1	2.7	52.5 ± 2.8	3.0	56.3 ± 0.3	5.8	55.4 ± 1.6	5.1	56.2 ± 0.8	5.9	96.1 ± 0.2
	10	84.1 ± 0.2	81.4 ± 3.3	2.8	70.3 ± 11.9	2.3	47.5 ± 0.3	3.2	58.4 ± 0.4	5.6	61.0 ± 6.5	5.6	62.1 ± 1.5	5.8	
	20	85.2 ± 0.6	83.5 ± 1.4	3.4	81.7 ± 1.9	2.2	68.2 ± 3.1	3.3	65.8 ± 3.4	5.8	73.5 ± 4.4	5.5	70.6 ± 4.9	5.8	
	Base acc:	85.8 ± 0.3	83.2 ± 2.0	4.3	82.7 ± 1.9	2.2	79.0 ± 2.7	3.1	76.7 ± 2.9	5.6	83.9 ± 1.2	5.1	77.4 ± 3.1	5.6	
	43.7%	86.5 ± 0.1	83.9 ± 2.0	3.8	84.2 ± 0.6	2.3	83.5 ± 1.4	3.4	85.7 ± 0.4	5.5	84.6 ± 1.5	5.1	80.8 ± 5.0	5.5	
Rotten tomatoes	5	82.2 ± 0.3	82.4 ± 0.6	4.5	72.6 ± 2.8	2.5	55.7 ± 2.8	3.8	69.8 ± 3.0	4.9	60.0 ± 1.2	6.4	70.4 ± 4.2	5.4	88.1 ± 0.3
	10	82.9 ± 0.2	81.8 ± 0.9	5.5	75.3 ± 2.8	2.3	63.7 ± 2.9	3.0	71.5 ± 2.9	5.2	61.1 ± 3.8	5.5	74.5 ± 4.1	5.8	
	20	84.4 ± 0.5	83.2 ± 0.6	7.0	78.0 ± 0.5	2.2	75.7 ± 2.4	3.1	79.1 ± 1.2	5.7	67.5 ± 0.7	5.2	80.6 ± 0.9	5.7	
	Base acc:	84.9 ± 0.2	83.1 ± 0.6	4.6	77.5 ± 0.2	2.3	78.7 ± 0.9	2.9	81.2 ± 0.7	5.6	78.7 ± 1.5	5.1	81.1 ± 1.8	5.6	
	65.8%	85.0 ± 0.3	83.2 ± 0.7	4.5	81.3 ± 1.0	2.3	82.3 ± 0.3	2.9	82.8 ± 1.2	5.6	82.1 ± 1.2	5.1	83.7 ± 1.1	5.6	

2013), Tweet emotions (Mohammad et al., 2018), and Rotten tomatoes (Pang & Lee, 2005b).

Model. We use the Phi model (Li et al., 2023a) to generate synthetic data and for supervised fine-tuning.

Fine-tuning settings. We fine-tune each model for 200 steps with Adam optimizer (Kingma, 2014) and batch size of 16. The learning rate follows a linear scheduler with the initial learning rate selected from $\{7e-6, 1e-5, 1.5e-5\}$. We run an evaluation every 50 steps and report the best test classification accuracy among all the checkpoints.

Baselines. We compare our method with LLM-generated synthetic data with zero-shot and few-shot methods (Li et al., 2023b). We also compare to popular coreset selection methods, namely Herding (Welling, 2009), K-center (Farahani & Hekmatfar, 2009), and Random.

Hyperparameters. The number of synthetic tokens is set to the average token length of all samples. For ADMM, the number of updates T is set to 30 and ρ is chosen from $\{0.001, 0.05, 0.01, \dots, 10\}$. To update X , we run 50 iterations of Adam with $\text{lr} = 0.008$. For the top- k projection, we use $k = 200$. For DP, we use $\delta = 1e-4$ and $\epsilon = 0.05$.

5.2. Main results

In our experiments, we consider the two scenarios discussed in Sec. 3. First, we apply GRADMM to generate synthetic training data based on a small number of examples from a target task. Then, we apply GRADMM to generate a small set of synthetic data by distilling an existing training data.

5.2.1. GENERATING LARGER SYNTHETIC FINE-TUNING DATA IN DATA-SCARCE REGIME

First, we consider the case where data is scarce for the target task, and we wish to generate a larger synthetic training data based on a small number of examples from the target task. Figure 1 shows the result of applying GRADMM to generate 100 synthetic examples based on only 5, 10, 20, 50 examples randomly selected from the validation data of SST-2, Tweet emotions, and Rotten tomatoes. We see that GRADMM successfully generates high-quality supervised fine-tuning data that can train Phi to a superior performance over that of training on the available validation data. Notably, GRADMM generated synthetic data based on only 5 real examples outperform the real data by 15.7%, 29.7%, and 31.5% on the three datasets. This confirms the effectiveness of GRADMM in the data-scarce regime.

5.2.2. GENERATING SMALL SYNTHETIC DATA BASED ON LARGER FINE-TUNING DATA

Next, we consider the case where a relatively large supervised fine-tuning data is available, and we generate a smaller synthetic data to replace the real data to preserve the privacy of training examples or to improve the training efficiency.

GRADMM outperforms baselines and preserves privacy.

Table 1 compares the performance of fine-tuning on synthetic data generated by GRADMM to that of zero-shot and few-shot techniques. It also shows the performance of fine-tuning on subsets of real data selected by herding, K-center, and Random baselines. We note that among all the methods, only the synthetic data generated by GRADMM can pre-

Table 2. Fine-tuning Llama-3.2-1B and OPT-1.3B on 20 synthetic samples generated by matching the gradient of a pretrained Phi.

Model	Dataset	Pretrained	GRADMM	Zero-shot	Few-shot	Herding	K-centers	Random real
Llama-3.2-1B	SST-2	68.6	89.4	82.4	79.7	85.4	64.6	<u>88.4</u>
	Tweet emotions	43.7	85.8	83.4	74.4	76.1	88.5	<u>83.9</u>
	Rotten tomatoes	67.5	87.8	80.5	78.5	73.6	87.8	<u>84.3</u>
OPT-1.3B	SST-2	62.3	<u>87.0</u>	83.9	85.6	85.8	73.8	88.7
	Tweet emotions	43.7	78.5	77.8	76.9	75.3	74.7	77.7
	Rotten tomatoes	63.1	87.9	74.9	80.6	80.8	84.8	<u>85.5</u>

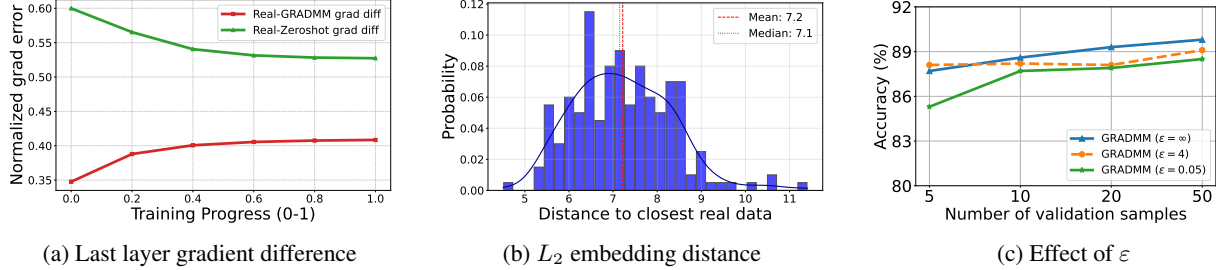


Figure 2. Fine-tuning Phi on synthetic texts generated by GRADMM for SST-2. (a) Normalized last-layer gradient error during fine-tuning on synthetic data. Data generated by GRADMM yields significantly smaller gradient errors compared to the zero-shot baseline, indicating closer alignment with real data. (b) L_2 embedding distance between GRADMM’s synthetic texts to their closest real training data. (c) Effect of ϵ on the performance of fine-tuning Phi on GRADMM’s synthetic data.

Table 3. SST-2. Divergence (FID) between the (i) training data distribution (Train), (ii) distribution of the few available real examples (Val), (iii) distribution of the 100 GRADMM synthetic data, (iv) distribution of 100 zero-shot synthetic data (Zero).

# data	(Train Val)	(Train GRADMM)	(Train Zero)
5	71.8	44.2	
10	59.8	43.3	
20	51.6	39.8	56.0
50	40.8	39.7	

serve the privacy of training data. We see that GRADMM outperforms all the baselines across various datasets and data sizes, by up to 13.1%. Notably, the synthetic data generated by GRADMM has a similar perplexity to that of real data, while having higher perplexity than LLM-generated synthetic data with zero-shot and few-shot methods. This confirms the more diverse nature of the synthetic data generated by GRADMM, compared to LLM generated data.

GRADMM’s synthetic data transfer to other LLMs. Table 2 shows the performance of fine-tuning Llama-3.2-1B and OPT-1.3B on 20 synthetic examples generated with GRADMM by matching gradient of a pretrained Phi model. We see that the data generated by GRADMM outperforms zero-shot and few-shot methods and the real data selected by herding and K-center baselines. This confirms the transferability of the synthetic data generated by GRADMM.

5.3. Analysis

GRADMM yields similar gradient to real data during fine-tuning. For fine-tune Phi on GRADMM’s synthetic data generated from SST-2, Figure 2a illustrates that the

normalized *last-layer* gradient error, i.e. $(\|\nabla_{\theta_L} \mathcal{L}(\theta_t) - \nabla_{\theta_L} \mathcal{L}^s(\theta_t)\|) / \|\nabla_{\theta_L} \mathcal{L}(\theta_t)\|$ at the pretrained parameters is small, and this relation holds during fine-tuning. Notably, GRADMM generated data has a much smaller gradient error than the zero-shot baseline during fine-tuning, corroborating its superior performance. Similar results for other datasets and *full* gradient error can be found in Appendix D.

GRADMM’s synthetic data is close to real data. Table 3 compares (for Figure 1a) the embedding divergence (in terms of FID) between the (i) training data distribution, (ii) the distribution of the few available real examples, (iii) the distribution of the 100 synthetic data generated by GRADMM and (iv) the distribution of 100 synthetic data generated using the zero-shot approach. Our synthetic data has a smaller FID, confirming that it has a more similar distribution to that of real training data, compared to baselines. This corroborates the superior performance of GRADMM. While the effectiveness of GRADMM depends on the diversity of the available real examples, our empirical results show that a small number of randomly selected examples can be leveraged to effectively reduce the expected loss.

GRADMM’s synthetic data is yet different from real data. Figure 2b shows the histogram of the distances of synthetic examples to their closest real training data. None of the synthetic examples generated by GRADMM are very similar to the real training examples, confirming that our synthetic data is not identical to real examples.

GRADMM preserves privacy. We apply loss-based MIA (Shokri et al., 2017) to the model fine-tuned on GRADMM’s synthetic data generated for SST-2 (Table

Table 4. SST-2. Matching the gradient of last-layer yields a higher performance with smaller number of synthetic data. Mapping the optimized embeddings to text via top- k projection (readable text) yields 9.2% higher accuracy than L_2 projection (unrelated words).

Method	Acc	#data	ppl
GRADMM	90.0	68	5.2
GRADMM with full grad	89.6	89	5.5
GRADMM w/o top- k projection	<u>80.8</u>	57	<u>13.3</u>

Table 5. SST-2. Our filtering strategies effectively reduce the size of synthetic data from 200 to 68 and yield 1.9% higher accuracy.

Method	Acc	#data	ppl
ADMM	88.1	200	4.6
+ Removing wrong labels	89.4	169	4.6
+ Selecting data with lowest loss	89.4	100	5.4
+ Balancing avg loss of categories	90.0	68	5.2

Positive: Great movie review is a must see experience that will leave you in a state of all time high with the brilliant acting and the stunning production.
Positive: The movie truly left me completely moved and in a better place than when I started it because of its well thought out and impactful way.
Negative: The overall quality of action in this movie was not impressive enough to keep me away from the action center.
Negative: Terribly bad and boring to me as a person who values quality content and a good storyline over mind.

Figure 3. Synthetic examples generated by GRADMM from SST-2.

1). We select $N = 100$ member samples from the training data and N non-member samples. Then, we find the optimal threshold that maximizes the advantages, defined as $2 \times (acc - 50\%)$, on these $2N$ samples. Finally, we test loss-based MIA with optimal threshold on another $2N$ samples consisting of N members and N non-members. Averaged advantage scores over 10 runs (smaller absolute values indicate better privacy) are $-2.5\% \pm 3.3$ for $\varepsilon = 0.05$ and $-2.9\% \pm 5.0$ for $\varepsilon = \infty$. We see that even our non-DP version retains strong privacy, performing only slightly worse than the explicitly differentially private version.

5.4. Ablation study

Generating readable text improves the performance. Table 4 shows that mapping the optimized embeddings to text via top- k projection yields readable synthetic text with low log-perplexity (ppl). In contrast, synthetic examples generated via L_2 projection have a considerably higher ppl, as they contain a set of unrelated words. Notably, the top- k projection yields 9.2% better performance than L_2 projection. This confirms that readability of the generated text is not

only important for interpretation and transferability of the results, but it is crucial for obtaining satisfactory performance.

Matching last-layer gradient boosts the performance, memory and speed of generation. Table 4 shows that matching the gradient of last-layer yields a higher performance with smaller number of synthetic data. At the same time, it reduces the generation memory by 2.6x (from 44.6G to 17.3G) and reduces the generation time by 2.3x (from 4.6 hours to 2 hours on one H100 GPU).

Filtering improves the performance of synthetic data.

Table 5 shows the effect of the three filtering strategies discussed in Sec. 4.4 to obtain a subset of at most $r = 100$ synthetic examples from the 200 synthetic data generated by ADMM. We observe that (i) removing examples that belong to the wrong category effectively improves the performance; (ii) selecting the top $r = 100$ examples with the lowest loss in every category effectively reduces the size of the synthetic data without harming its performance; (iii) dropping examples with highest loss in categories that have a larger average loss further reduces the size of the synthetic data while improving its performance. The filtering strategies reduce the size of the synthetic data from 200 to 68, while yielding 1.9% improvement in the fine-tuning performance.

Effect of ε . Figure 2c compares the performance of synthetic texts generated with different ε values. As expected, increasing ε , i.e., relaxing the privacy constraint, generally leads to better performance. Interestingly, when the number of validation samples is limited to just 5, the DP version with $\varepsilon = 4$ outperforms the non-DP version ($\varepsilon = \infty$). We hypothesize that the injected noise may enhance generalization during training, consistent with observations from prior work (He et al., 2021).

Qualitative results. Fig. 3 shows examples of generated synthetic text by GRADMM from positive and negative classes of the SST-2. We see that the synthetic data is meaningful and semantically aligns with the target categories.

6. Conclusion

We proposed the first theoretically-rigorous method for generating privacy-preserving synthetic readable text data by matching the gradient of real examples from a target task. We formulated this problem as a discretely constrained non-convex optimization in the embedding space and applied the Alternating Direction Method of Multipliers (ADMM) to iteratively optimizes the embeddings of synthetic examples to match the noisy target gradient, and map them to a sequence of text tokens with low perplexity. We proved that the generated synthetic text can guarantee convergence of the model to a close neighborhood of the solution obtained by fine-tuning on real data. Our experiments on various classification tasks confirmed the effectiveness of GRADMM.